

Remaining Implementation Tasks

CRITICAL

C-01: B2 and Resend credentials not configured

The entire B2 storage layer and all transactional emails are broken in the running environment. The code is correct – the secrets just aren't set.

Missing secrets:

- RESEND_API_KEY – all verification, welcome, approval, reset, and storage alert emails are silently skipped
- B2_APPLICATION_KEY_ID, B2_APPLICATION_KEY, B2_BUCKET_NAME – all file uploads and downloads throw credential errors

Action: Set these four secrets via the Secrets panel. The app will fully enable without any code change.

C-02: Offline-First Architecture – Not implemented at all

The SRS defines this as a mandatory constraint (C-07) and a high-priority NFR. Currently the exam engine has zero offline resilience.

What's missing:

- **Exam answers are not persisted to IndexedDB** – if the browser crashes mid-exam, all answers are lost

- **No reconnect/sync logic** – SRS AC-EX27 requires auto-submit after 10 min disconnection
- **No service worker** – the app is not installable as a PWA and has no cache layer

Files to create:

- `artifacts/edtech-platform/public/sw.js` – Service Worker (cache-first for assets, network-first for API)
- `artifacts/edtech-platform/src/lib/exam-offline-store.ts` – Dexie.js wrapper for answer persistence
- `artifacts/edtech-platform/src/lib/offline-sync.ts` – reconnect detection + auto-submit trigger
- `artifacts/edtech-platform/public/manifest.json` – PWA manifest (icons, name, display: standalone)

File to modify:

- `artifacts/edtech-platform/src/pages/exam-interface.tsx` – integrate Dexie writes on every `handleSaveCurrentAnswer` call; add `navigator.onLine` disconnect timer

C-03: QR Code Auto-Generation not wired to Admin Question Creation

The database has `qr_code_svg` on the questions table, and the results page renders it – but the admin question creation/edit flow never generates the QR code. Any question saved with a `videoUrl` should auto-generate an SVG QR code pointing to that URL at save time.

File to modify:

- `artifacts/api-server/src/routes/questions.ts` – on `POST /questions` and `PATCH /questions/:id`, if `videoUrl` is set, call `qrcode.toString(videoUrl, { type: 'svg' })` and persist to `qrCodeSvg`

 HIGH

H-01: Exam Timer Server-Side Sync – Manipulation vector remains open

SRS constraint C-13 requires server-client timer sync to prevent cheating. Currently remainingSeconds is only written to the DB on pause and resume. A user who manipulates Date.now() on the client can effectively bypass the countdown.

The fix is lightweight – no websocket needed:

File to modify:

- artifacts/api-server/src/routes/exams.ts – add GET /attempts/:attemptId/sync-time endpoint that calculates remainingSeconds server-side as storedRemaining - (now - resumedAt) and returns it
 - artifacts/edtech-platform/src/pages/exam-interface.tsx – sync against this endpoint every 60 seconds, hard-set timeLeft to the server value if drift > 5 seconds
-

H-02: Exam Keyboard Shortcuts – Missing

SRS EX-08 specifies Arrow keys for navigation, M to toggle mark-for-review, Enter to save-and-next. Not implemented.

File to modify:

- artifacts/edtech-platform/src/pages/exam-interface.tsx – add useEffect with keydown listener; gate shortcuts to not fire when inside <input> or <textarea>

H-03: Lecture CTR Admin Analytics – Not surfaced in UI

The backend tracks `lectureClickCount` per user per topic. The `low_ctr_threshold` config key exists. The `/admin/qr-analytics` endpoint exists. But the admin dashboard page has no UI section showing which lectures have below-threshold click-through rates.

SRS AC-T05 and AC-T06 require this to be visible.

Files to modify:

- `artifacts/edtech-platform/src/pages/admin-dashboard.tsx` – add a "Low-CTR Lectures" card that queries the analytics endpoint and flags topics where `clickCount / enrolledStudents < threshold`
- `artifacts/api-server/src/routes/admin.ts` – extend `/admin/qr-analytics` (or add `/admin/lecture-ctr`) to aggregate click counts per topic against enrolled user count

H-04: Profile Photo Upload – Client-Side Compression Missing

The backend endpoint is correct. The `profile.tsx` page has the upload flow. However, SRS AC-PROF03 requires client-side compression to max 500KB and resize to max 500×500px **before** the file is PUT to the B2 signed URL. Without this, large photos waste B2 quota and violate the spec.

File to modify:

- `artifacts/edtech-platform/src/pages/profile.tsx` – before calling `getUploadUrl`, use the browser's Canvas API to resize and compress the image: `createImageBitmap` → draw to 500×500 canvas → `canvas.toBlob('image/jpeg', 0.85)` → upload that blob instead of the original file

H-05: Multi-Subject and Grand Test – No Entry Point in Frontend

The backend fully supports `subject_test` and `grand_test` exam types with correct gating. However, the subjects list page and subject detail page have no section showing these tests. Students cannot see or access them from the UI.

Files to modify:

- `artifacts/edtech-platform/src/pages/subject-detail.tsx` – add a "Subject Test" section at the bottom, shown with lock/unlock state based on `isUnlocked` from the exams query filtered by `type=subject_test`
 - `artifacts/edtech-platform/src/pages/subjects.tsx` – add a "Grand Test" card at the bottom, locked until all subjects complete
-

H-06: Admin Bulk Question CSV Import

SRS EX-07 and AC-EX22 require bulk question upload. The `import-syllabus` endpoint handles hierarchy (subjects/chapters/topics) but not questions. There is no `POST /admin/questions/bulk-import` endpoint.

Files to create/modify:

- `artifacts/api-server/src/routes/admin.ts` – add `POST /admin/questions/bulk-import` that accepts CSV with columns: `text`, `option_1..4`, `correct_option`, `topic_id`, `marks`, `text_solution`, `video_url`, `difficulty`; validates topic existence; inserts in batch; auto-generates QR if `video_url` present
- `artifacts/edtech-platform/src/pages/admin-questions.tsx` – add a CSV upload button/modal with format documentation

 MEDIUM

M-01: Web Push Notifications – Schema exists, code doesn't

The `push_subscriptions` table exists. In-app notifications (DB) are fully working. But the SRS FR-NOT-01 specifies a third channel: browser Web Push (reminders like "Your Daily Plan is ready", "Streak about to break"). Nothing implements the `PushManager.subscribe()` flow, the VAPID key generation, or the server-side push dispatch.

Files to create:

- `artifacts/edtech-platform/src/lib/push-manager.ts` – subscribe to push, store subscription via `POST /api/push/subscribe`
- `artifacts/api-server/src/routes/notifications.ts` – add `POST /api/push/subscribe` (save to `push_subscriptions`); add `sendPushNotification(userId, title, body)` helper using `web-push` npm package

Secrets needed: `VAPID_PUBLIC_KEY`, `VAPID_PRIVATE_KEY` (generate once with `web-push generate-vapid-keys`)

M-02: Password Strength Meter UI – Missing

SRS FR-AUTH-01 requires a visual strength meter during registration and password change. The backend enforces the policy but there's no client-side strength feedback.

File to modify:

- `artifacts/edtech-platform/src/pages/register.tsx` – add real-time password strength indicator (e.g., a colored progress bar: Weak / Fair / Strong / Very Strong) based on length, uppercase, number, special char checks
 - `artifacts/edtech-platform/src/pages/profile.tsx` (password change section) – same indicator
-

M-03: PDF Annotations – Not implemented

SRS FR-P02 / AC-P07 requires highlights and notes stored as JSON and persisted across sessions. The `inline_notes` table handles plain text per topic but there is no annotation layer for uploaded PDFs.

Files to create:

- `artifacts/edtech-platform/src/lib/pdf-annotations.ts` – data model for highlight ranges and note anchors; persist to server via a new endpoint
 - `artifacts/api-server/src/routes/notes.ts` – add GET/PUT `/notes/:noteId/annotations` endpoints (store as JSON blob in a new `note_annotations` column or table)
-

M-04: PDF Full-Text Search – Not implemented

SRS AC-P05 requires text extracted from PDFs to be searchable. Currently notes are stored and retrieved by filename only.

Files to modify:

- `artifacts/edtech-platform/src/pages/notes.tsx` – after download URL is fetched, use `pdf.js` (`pdfjs-dist`) to extract text in-browser and store in `localStorage` keyed by note ID

- Add a search input to the notes page that filters against extracted text
-

M-05: Notification Preferences UI – Missing

SRS FR-NOT-02 requires toggleable preferences for Daily Plan and Streak reminders. The notifications table and backend exist but there's no settings UI.

File to modify:

- artifacts/edtech-platform/src/pages/admin-settings.tsx or a new user settings.tsx page – add a "Notification Preferences" section with toggles persisted to a notification_preferences field on the users table or a separate table
-

M-06: Exam Result Percentile – Always null

SRS EX-03 lists percentile as a result metric. The results endpoint returns percentile: null in all cases. The calculation is straightforward: (count of users with lower score on same exam / total users who attempted) * 100.

File to modify:

- artifacts/api-server/src/routes/exams.ts – in the submit result handler, after inserting the result, query all results for the same examId, compute percentile rank, and persist it
-

M-07: Emergency Super-Admin Recovery – Not implemented

SRS FR-B00T-03 requires a recovery mechanism for lost super-admin access. Currently, if the sole super-admin account is compromised or deleted, there is no recovery path short of direct DB access.

File to create:

- artifacts/api-server/src/routes/admin.ts – add POST /api/admin/emergency-recovery gated by a RECOVERY_SECRET env var (checked via constant-time compare); promotes a specified userId to super_admin

Secret needed: RECOVERY_SECRET (set once, stored offline by the founder)

 LOW

L-01: Cloudflare CDN in Front of B2 – Not configured

SRS AC-P09 requires CDN caching to reduce B2 egress by >80%. Currently the signed URLs point directly to the B2 endpoint (s3.us-east-005.backblazeb2.com). This is an infrastructure-level task, not a code change, but it's a spec requirement.

Action: Create a Cloudflare Worker or set up a Cloudflare custom domain that proxies B2 bucket URLs; update B2_ENDPOINT in artifacts/api-server/src/lib/b2.ts to point to the CF domain for download URLs.

L-02: "Focus Master" Badge Gamification – Not implemented

SRS FR-POM0-02 specifies a "Focus Master" badge awarded after a 7-day Pomodoro streak. The streak is calculated but nothing awards or displays a badge.

Files to modify:

- artifacts/api-server/src/routes/pomodoro.ts – after logging a session, check if streak ≥ 7 and call createNotification with a badge-awarded message
 - artifacts/edtech-platform/src/pages/dashboard.tsx – display earned badges
-

L-03: Custom Pomodoro Mode – Missing

SRS FR-POM0-01 specifies Standard (25/5), Long Break (15min), and Custom modes. The frontend component likely only has fixed modes – Custom duration input is not implemented.

File to modify:

- artifacts/edtech-platform/src/components/pomodoro.tsx – add a "Custom" option that reveals duration inputs for work and break periods
-

L-04: Sentry Error Monitoring – Not integrated

SRS NFR-REL-02 specifies 100% of errors logged to Sentry. Currently errors are logged via pino (server) and unhandled exceptions crash silently client-side.

Files to modify:

- artifacts/api-server/src/lib/logger.ts – add @sentry/node init in production

- artifacts/edtech-platform/src/main.tsx – add @sentry/react init with ErrorBoundary

Secret needed: SENTRY_DSN

Implementation Roadmap

The logical order to stay unblocked and shippable at each step:

Phase 1 – Unblock the infrastructure (1-2 hours, no code required)

└ C-01: Set B2 and Resend secrets → immediately enables emails + file uploads

Phase 2 – Security and integrity (critical before any users)

└ C-03: QR auto-generation on question save

└ H-01: Server-side timer sync endpoint

└ M-07: Emergency recovery secret

Phase 3 – UX completeness (high-visibility missing features)

└ H-02: Exam keyboard shortcuts

└ H-04: Profile photo client-side compression └ H-05: Subject test + Grand test entry points in frontend

└ M-02: Password strength meter

└ M-06: Percentile calculation

Phase 4 – Offline resilience (required for mobile users)

└ C-02: Service worker + IndexedDB exam answer persistence

Phase 5 – Admin tooling

- └─ H-03: Lecture CTR analytics dashboard section

- └─ H-06: Bulk question CSV import

Phase 6 – Advanced features

- └─ M-01: Web push notifications

- └─ M-03: PDF annotations

- └─ M-04: PDF text search

- └─ M-05: Notification preference UI

Phase 7 – Production hardening

- └─ L-01: Cloudflare CDN for B2

- └─ L-04: Sentry integration

- └─ L-02: Focus Master badge

- └─ L-03: Custom Pomodoro mode

The only item that blocks all other testing right now is **C-01** – set the four secrets and the majority of the platform becomes fully functional.